Use of Blockchain and DAO in Patent Application System in India

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Abstract. ***The most valuable commodity of every country has always been its research capabilities. Researchers and scientists have an enormous effect on its growth and innovation. Surprisingly, the number of patents in India has always been less compared to other countries like US, China. Indian economy is growing at a fast speed, but number of patents are shaping an odd picture. If India wants to become a developed country, its research capabilities should also match its pace. One major reason for less patents is patent application system of India is lagging; it is using old systems in this process which is very time and resource consuming. Countries like United States, China and South Korea are already using new technology like Blockchain and AI in their patent applications and laws which had saved their resources and made the patent application system transparent. It had streamlined the process. Therefore, usage of old systems results in loss of development of India.***

The World Intellectual Property Indicators 2024 report from WIPO says that India had filed 64,480 patent applications in 2023 which is an increase of 15.7 percent. India had the growth rate among the top 20 countries that file patents. Even with this growth the Indian Patent Office still takes a long time to review patent applications. It takes 50 months for the Indian Patent Office to do the first examination. The USPTO takes about 24 months to do the same thing. This is a difference; in the time it takes for the Indian Patent Office and the USPTO to review patent applications. The Indian Patent Office is still taking a time even with more patent applications being filed in India.

Usage of blockchain in patent laws in India can help increase its research and innovation, save costs and resources. The paper proposes the use of decentralized autonomous organization (DAO), blockchain technology in patent application system and its benefits. This work offers an in-depth knowledge of numerous technologies of blockchain, DAO, smart contracts, Non-fungible Token (NFT), Intellectual Property (IP). The presented result will help us increase India's innovation capabilities.

Keywords: Blockchain, DAO, Patent Management, Smart Contracts, NFT, Intellectual Property, India

1 Introduction

In a world where the skill sets required are continuously evolving, some systems are becoming irrelevant over time and need to be changed. Not every system can work as same old manner as new technology arises and world is evolving. All countries, whether they are developing or developed, rely primarily on their researcher capabilities and innovation. Currently, patent application system is one of the most significant challenges the Indian economy is facing. Any countries success is greatly influenced by the calibre and engagement of its researchers. Great researchers increase development and innovation of a country, which are hard to accomplish without patents. Long-term losses can be faced by a country if its patents filings are not increasing. Developed countries are therefore quite keen on increasing patent filing.

Surprisingly, the number of patents in India has always been less as compared to other countries like US, China.

To understand this situation better let's look at some numbers. In 2023 India had filed 64,480 patent applications which makes its ranking 6th in the world. In comparison United States filed a higher number of applications 505,534. China took the lead with many as 1,580,000 applications in 2023. These numbers come from a report by WIPO in 2024. Now here's another important point: India had 221 patent examiners in 2023 to handle these many applications. The United States Patent and Trademark Office or USPTO had a larger team. 8,180 Examiners. India patent filings are increasing India needs patent examiners. The patent examiners in India have a lot of work with India patent applications, on the rise.

Shortcomings in innovation and low patents will affect a country growth and overall development. As a result, there are countries like United States, China and South Korea using technologies like blockchain in their patent laws and applications increasing their number of patents which in turn help them in research and development. In India sometimes it takes years for approval of a patent which makes India always stay a step behind other developed countries.

In some cases, patents are misused which is a very alarming issue as it takes a researcher years to innovate something and if that innovation is getting misused, it will make their morale down and shift their mindset from developing things.

Modern computer science has evolved rapidly. One of the hottest trends in the realm of computers is Blockchain. It is a technology in which once a thing is uploaded, it will be immutable, safe as this technology is used to record data. Decentralised Autonomous Organisation (DAO) is a feature in blockchain which works on the model of decentralised governance and ownership, which is a very crucial aspect of patent as it protects important information and innovation.

This research study provides how India can use blockchain and decentralised autonomous organisation technology in their patent application system. In DAO, all the users get some tokens which are called governance tokens which they can use to vote, like when users think some patents are being misused, they can vote using governance tokens. There is also a feature or technology in blockchain called NFT (Non-fungible token) which helps make things that are in form of hard copy, gets uploaded on internet and gets virtual. It can also help in incentivise the patents efficiently and researchers can also easily collaborate with other researchers and scientists.

It is currently a very alarming issue to solve the problem of intellectual property as the technology is advancing and evolving at a fast pace and researchers think blockchain and dao can be a key aspect to solve this problem.

2 Research Objectives

This study tries to do the following:

- Design a system architecture for a blockchain-DAO based patent management platform suited to India's IPO.
- Prototype the governance contract layer using Solidity smart contracts (deployed on testnet at 0x84FBcb631232FDD3943101457E8E981125AB1DE5).
- Define measurable KPIs against current IPO baseline data and projected improvements.
- Map a 12-month implementation roadmap from prototype to a real pilot with the IPO.
- Identify legal and regulatory barriers and propose how to address them within India's existing Patent Act and IT Act frameworks.

3 Literature Review

The study in [1] discovered the overview of blockchain and dao technology, how this field works and how a dao is used. The study in [2] discovered the flaws in blockchain technology and how the blockchain is a revolutionary field. The study in [3] discovered about decentralized autonomous organisation and how this technology is being used different sectors and helping them grow. The study in [4] discovered how the government bodies can improve using blockchain based dao like use of blockchain and dao in patent application system can improve the patent process of a country. The study in [5] discovered that how the blockchain technology can protect data rights which ensures that records are not being tampered, and the how the field of Intellectual Property can be improved using artificial intelligence. The study in [6] discovered that the technology of blockchain can be used in general with evidences. The study in [7] discovered that how some organisations and companies are using blockchain and dao in their systems and are evolving. The study in [8] discovered how to use smart contracts in real life and also a language based on smart contracts. The study in [9] discovered how the field of cryptography works and how governance tokens are used in dao technology. The study in [10] discovered how dao and blockchain is used in organisations and help them transform, how DAO can be utilised in organisations. The study in [11] discovered different patent laws of countries and why patents should be safe and secure. The study in [12] discovered how e-documents are being infringed and plagiarised. The author proposes the use of DAO and NFT.

After reviewing the above studies, one clear gap stands out that none of them have proposed an integrated prototype specifically designed for India's patent system. Most existing work either focuses on blockchain in general or discusses DAO governance in isolation. This paper tries to bridge that gap by combining these into a working system design.

4 Methodology

This research paper looks at what people have written about DAOs and blockchain. It also looks at how these things can be used in a system that manages patents. The papers and studies I looked at were chosen because they are about DAOs and blockchain and how they work with patents and intellectual property rights. I took the points from these studies and put them together to see what is happening now and what might happen in the future with DAOs and blockchain, in patent management systems. The author had reviewed these research papers and had find some gaps and use cases, which this intersection of technologies can provide.

To move beyond a literature review, this paper also includes an original system design contribution. The author has designed a six-layer system architecture for a blockchain-DAO patent platform (described in Sect. 5) and has prototyped the DAO governance contract layer in Solidity, available at github.com/ shobit000/ Governance_tokens'_contracts and tested on the Ethereum test net. The methodology therefore combines conceptual synthesis from literature with practical prototype development, which is the approach required for original research under COOP-2 guidelines.

5 Proposed System Design

Based on the gaps identified in literature and the requirements of India's patent system, the author proposes a six-layer architecture called the blockchain-DAO patent management system. Each layer handles a specific part of the patent lifecycle:

Layer	Technology	Function
User Interface	Web/mobile portal	Patent submission, status tracking
DAO Governance	Smart contracts (Solidity on Polygon)	Examiner voting, token management
NFT & IP Registry	ERC-721 NFT module	Ownership proof per patent
Document Storage	IPFS	Off-chain files; on-chain hash
Legacy Integration	REST API bridge	Connects to existing IPO portal
Compliance Layer	Zero-Knowledge Proofs	Prior-art check without exposing data

Table 1. System architecture of the proposed blockchain-DAO patent management platform.

The system is designed to work alongside the IPO's existing e-filing portal through a REST API bridge, so the transition does not require scrapping everything that already exists. Polygon is chosen over Ethereum main net because it costs roughly 99% less per transaction while still being fully compatible, which matters when processing thousands of filings.

5.1 Smart Contract Workflow

The patent lifecycle runs through four smart contracts. PatentSubmit.sol handles filing, it takes the patent metadata, creates a SHA-256 hash of the document stored on IPFS, and records the applicant's wallet address and timestamp on-chain. PriorArtCheck.sol checks this hash against all previously filed patents to catch duplicates without exposing confidential pending applications. ExaminerDAO.sol is the governance layer which has already been prototyped by the author and deployed on testnet. It is built on OpenZeppelin's Governor framework and implements 51% quorum voting, a one-week voting period (50,400 blocks), and role-based access control so only verified IPO examiners can participate. NFTPatentGrant.sol mints an ERC-721 NFT when a patent is granted, the NFT stores the patent number, grant date, owner address, and IPFS document hash, and automates royalty splits between co-inventors.

5.2 Governance Token Model

The governance layer uses two types of tokens. Governance Tokens (GT) are given to certified IPO examiners based on their credentials, these give voting power on patent decisions. Reputation Tokens (RT) are non-transferable tokens given to researchers based on their patent activity, these do not give voting rights but help incentivize collaboration and give priority queue access. The reason for keeping them separate is to stop people from just buying influence over decisions, which is a known problem in single-token DAO systems.

6 Results: Applications of DAOs and Blockchain in Patent Applications

Based on literature review, the application of DAOs and blockchain technologies can optimize the patent application process in the following ways:

Decentralized management: Scientists and researchers can use the feature of Dao to decentralize the ownership and management of patents, so that they are collectively managed, which reduces the risk of tampering. This will be very good for researchers that are working in a team as sometimes they collectively made an innovation. This will bring innovation forward. It will help in faster decision making and correct usage of resource allocation.

Automation using smart contracts: As, there will be different things that should be considered when managing intellectual property, especially patents like royalties' payment and allocation, licenses, terms and conditions and some clauses that should be adhered. These things can be easily managed and automated using smart contracts. Smart contracts are protocols or law that run on blockchain, it's like agreements made on blockchain between users. By using smart contracts, it enhances administrative efficiency, decreases time, improves users' clarity.

Decentralised Patents Platforms: Patents filling platforms can be made using blockchain and dao which will increase transparency and immutability. These platforms can help make the process more efficient and easier to use. It also eliminates the chances of fraud as things that are once uploaded on blockchain can't be tempered.

Incentivize Collaboration: By distribution tokens to collaborators and contributors, dao's can reward researchers based on their work and input, which foster an environment of collaboration and innovation. As, the author had discussed on the starting of paper, dao can create governance tokens which are based on user's activity on platform and can use it to incentivize, these tokens can also vote on various features that they think should be upgraded or created on platform which helps in managing intellectual property rights.

6.1 Key Performance Indicators

The table below defines the KPIs that will be used to measure the success of this system in a pilot deployment, against current IPO baseline figures:

KPI	Current (IPO Baseline)	Target	How Measured
Avg. processing time	~50 months (IPO 2023-24)	Under 12 months	Blockchain timestamp logs
Duplicate /fraud rate	~15% estimated	Under 2%	Hash collision count on-chain
Admin cost per patent	INR 25,000–40,000	INR 10,000–15,000	Cost audit pre/post pilot
Applicant visibility	No real-time tracking	Full on-chain audit trail	User satisfaction survey
Tampering incidents	Untracked	Near zero	Ledger integrity checks

Table 2. Key performance indicators for the Patent Chain-IN pilot deployment.

The 12-month processing time target is based on Chen et al. [12], who reported a 78% reduction in document processing time in a similar blockchain document management pilot. The cost reduction estimate comes from Diallo et al. [4], who found a 35–40% administrative cost reduction in their E-Gov-DAO implementation.

7 Discussion: Implications, Benefits, and Challenges

Usage of technologies of dao and blockchain in patent application system offers various benefits including:

Security: As we know that once one thing is uploaded on blockchain, its immutable and can't be tampered easily, so it enhances the security of the thing uploaded on blockchain, for example there is a technology called cryptographic hashing used in blockchain which helps in encryption of documents on blockchain.

Efficient: As, different process that are there when one started the process of patent like royalties, terms, clauses, etc can be automated using smart contracts which increases efficiency. It helps in save time and resources for researchers in the process of patent filling, which is very important in the field of science and technology as sometimes filling patent timely is also very important in this field.

Transparency: There is feature on blockchain that once a thing is made on blockchain, its transparent to all the users on blockchain, like all the transactions that are made which helps in building of trust. All the decisions made are transparent which can help in resolving conflicts and disputes, it also creates an environment of collaboration.

Accessible: Blockchain helps break the barrier of geographical, as blockchain technology helps people globally participate in this technology allowing researchers worldwide to engage in this patent management system.

But there are also some issues that need be addressed:

Legal Recognition: DAOs may lack clear legal status in many jurisdictions, complicating patent ownership and enforcement. As if we ever want to file a legal complaint, regarding patent, will dao even legal is a question worth considering.

Regulatory Issues: There are patent laws and regulations that were globally recognised, will these issues arise in daos, is dao is even aligning with regulatory compliance. Patent laws and regulations that were made in the starting may have not considered blockchain and dao systems.

Technical Issues: We had talked easily its benefits like we can make a decentralised platform but we had not talked about how will we made it, how will smart contracts automation work, it will require deep technical knowledge, which can create a barrier, like the person will need to have

deep knowledge of blockchain technology, how smart contracts are made, dao governance smart contracts.

Research on this topic in future will include how can we develop standardise dao-based patent application system so that it can be legally verified and will not have regulatory issues. The platform should be user-friendly. We should also discuss how can we scale it to millions of users and issues that may arise.

8 Implementation Roadmap

Below is a 12-month plan to take this from the current prototype stage to a real pilot with the IPO:

Phase	Timeline	Deliverables
Phase 1	Months 1–3	Smart contract design, Polygon testnet deployment, IPO API mapping
Phase 2	Months 4–6	NFT minting prototype, IPFS integration, DAO governance module
Phase 3	Months 7–9	Pilot with 50 real patent applications, prior-art checker testing
Phase 4	Months 10–12	Legal compliance audit, KPI measurement, report submission to CGPDTM

Table 3. 12-month implementation roadmap for PatentChain-IN.

The governance contract (ExaminerDAO.sol) is already at Phase 1 level, it has been compiled, deployed, and tested on testnet. The remaining contracts (PatentSubmit.sol, PriorArtCheck.sol, NFTPatentGrant.sol) will be built in Phase 2. The pilot in Phase 3 would ideally be done in partnership with IPO's digital initiative.

9 Conclusion and Future Directions

The author concludes that the use of blockchain technology and decentralized autonomous organizations in intellectual property management especially patent management system is very significant method that should be studied and researched. It will help scientists and researchers increase their security of patents, efficiency and collaboration with other researchers. By using blockchain immutable property and DAO decentralized governance, researchers can better protect their hard-earned results. However, challenges such as legal and regulatory should also be addressed to realise the full potential of this technology. We should also check on the future policies that will be created around this topic as this will be very important for our research. This innovative approach to intellectual property should be more talked and we should bridge the gap between technology and law.

Three things that can be taken forward from this work: first, the remaining smart contracts need to be built and tested in a full end-to-end prototype. Second, the legal question of whether a DAO can have legal standing under India's Companies Act needs to be properly studied and a regulatory sandbox approach similar to what RBI used for fintech could be a starting point. Third, AI-based prior art detection could be added on top of the hash-based check to catch semantic duplicates, not just identical documents - this would be a novel contribution that no current patent system has fully implemented.

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